WEB APPLICATION ARCHITECTURE – AN OVERVIEW

PROGRAMMING APPLICATIONS AND FRAMEWORKS (IT3030)

LEARNING OUTCOMES

- After completing this lecture, you will be able to,
  - Describe the main components of three tier architecture for web development.
  - Identify some suitable technologies to develop the presentation layer and the application layer of a web application.
  - Apply suitable architectural components to design the presentation layer and the application layer of a Three tier web application.

CONTENTS

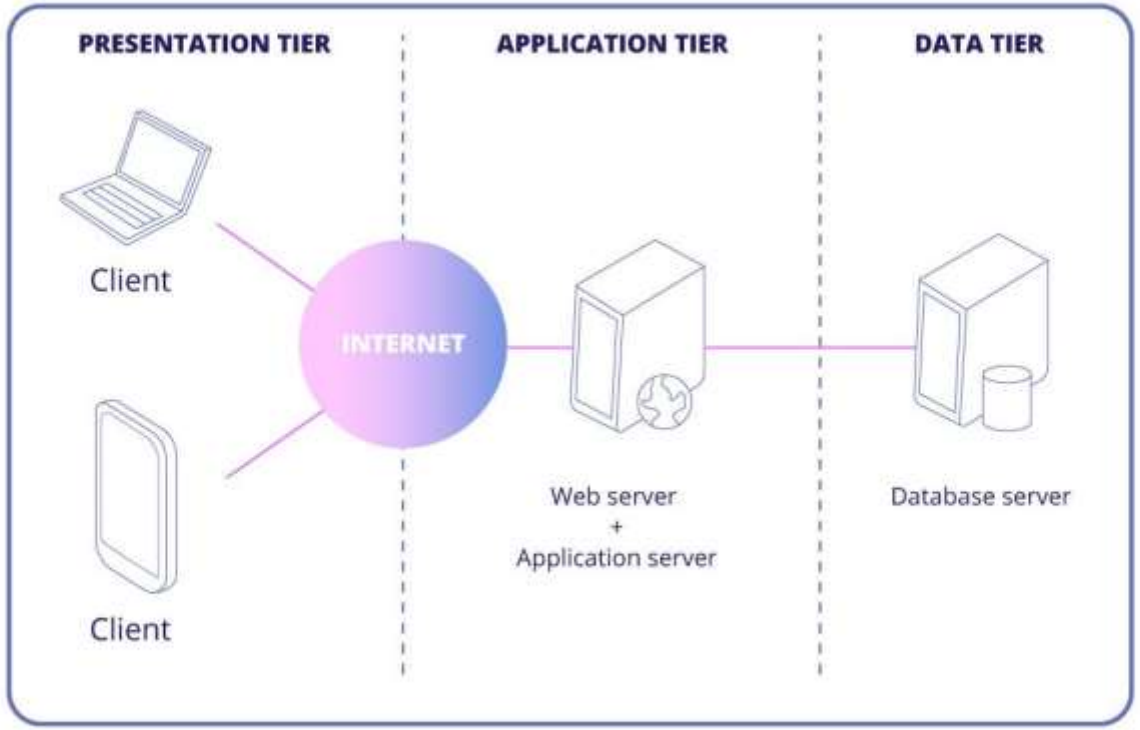
- Website vs Web Application
- Three Tier/ Three Layer Architecture
- Presentation Layer
  - Some technologies
  - Architecting the Presentation Layer
- Application Layer
  - Some technologies
  - Application Programming Interface (API)
  - Architecting the Application Layer
  - Big Ball of Mud: an architectural anti-pattern
- Other Helpful Technologies

WEBSITE VS. WEB APPLICATION

- What is the difference between a Website and a Web Application?
  - A Website provides static content which can be consumed by the end users.
  - A **Web Application** is designed for **interaction** with the end users.

- Our focus is on **Web Applications**.

THREE TIER ARCHITECTURE



Source: https://mobidev.biz/blog/web-application-architecture-types

mobidev

THREE TIER ARCHITECTURE

- Modern web applications are engineered based on this architecture.

- Has three tiers.
  - Presentation tier/ layer
  - Application tier/ layer
  - Database tier/ layer

- The most popular form of n-tier (multitier) architecture.

THREE TIER ARCHITECTURE

- Three tier architecture is a good choice for the development of web applications.

- Reasons?
  - Each tier/ layer runs on its own infrastructure.
  - Each tier/ layer can be developed independently/ in parallel by different teams.
  - Allows updating and scaling up/ down each tier independently without impacting the other tiers.
  - Better security when compared with two tier architecture (due to the isolation of data and logic from the presentation tier).

PRESENTATION LAYER

- Also called the client side or the frontend.
- Technologies used to implement include,
  - HTML/ CSS/ JavaScript
  - React.js
  - Angular
  - Vue.js
  - Some legacy technologies
    - jQuery and AJAX

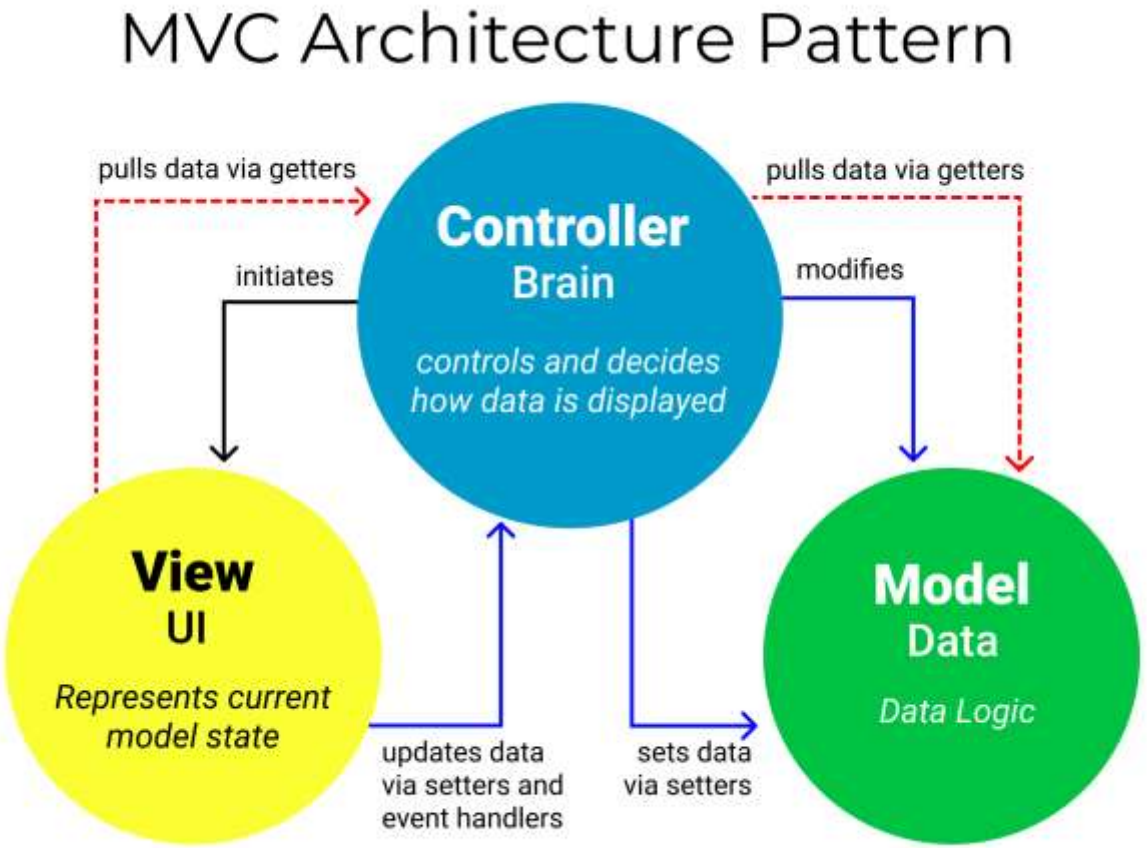
ARCHITECTING THE PRESENTATION LAYER

- Some well known examples
  - Model-view-controller (MVC)
  - Model-view-viewmodel (MVVM)

- Some more examples
  - Model-view-presenter (MVP)
  - Component Architecture
  - Micro frontends

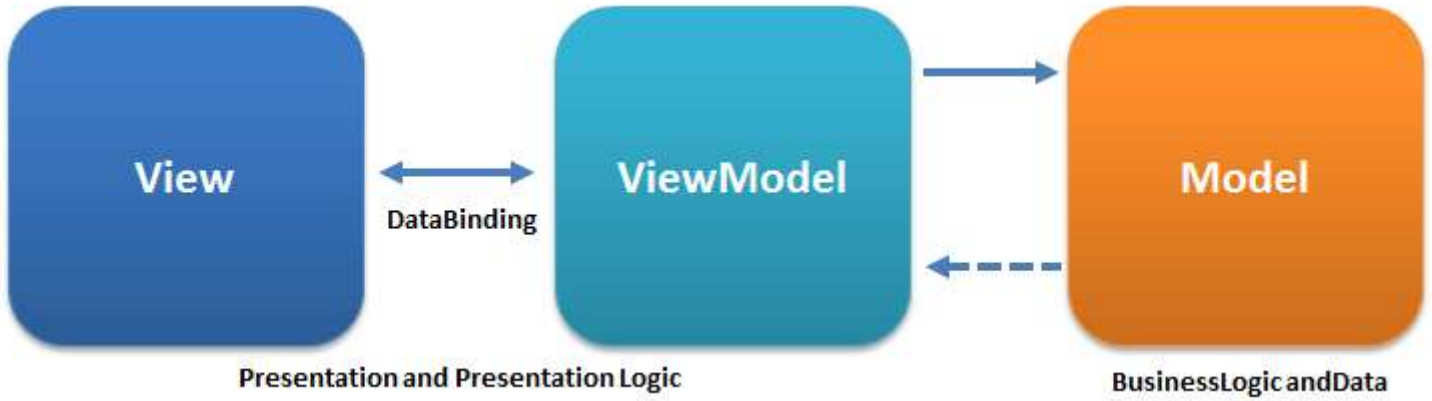
ARCHITECTING THE PRESENTATION LAYER: MVC

- [Youtube] MVC Explained



Source: https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/

ARCHITECTING THE PRESENTATION LAYER: MVVM



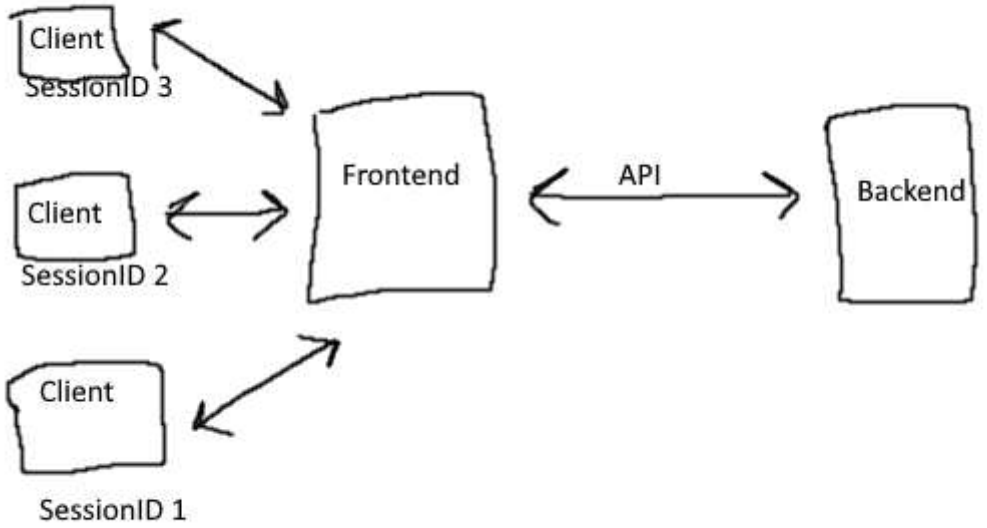
Source: https://en.wikipedia.org/wiki/Model%E2%80%93view%E2%80%93viewmodel

APPLICATION LAYER

- Also called the server side or the backend.
- Technologies used to implement include but not limited to,
  - Programming languages and frameworks
    - Java (frameworks: Spring/ Spring Boot)
    - Node.js (frameworks: express.js/ Koa)
    - Python (frameworks: Django/ Flask)
    - PHP (frameworks: Laravel, Symphony)
    - .Net Framework
  - Virtual Machines (VMs)
  - Serverless Computing
  - Containers

APPLICATION LAYER

- How does the Backend and the frontend communicate?



Source: https://stackoverflow.com/q/71033687

APPLICATION LAYER: APIS

- Application Programming Interface (API)!

- What is it?
  - A software interface which facilitates communication between two or more applications.
  - Abstracts away the complexities of application integrations.
  - A contract of services offered.

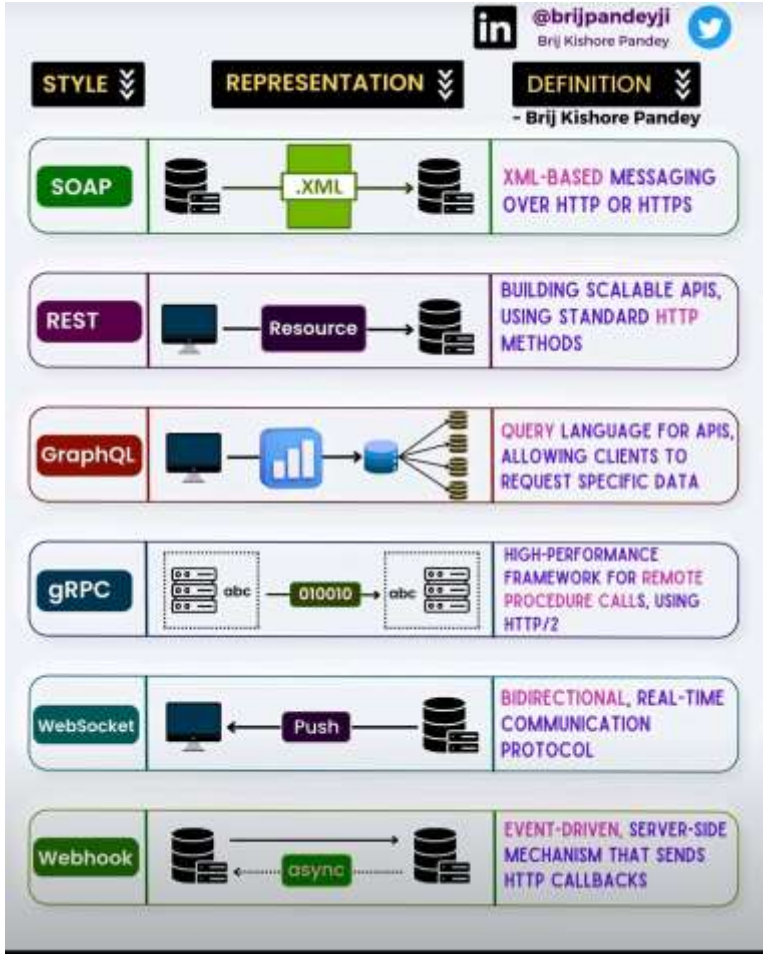
APPLICATION LAYER: APIS

- Benefits of APIs?
  - developers don't have to create everything from scratch - can use existing functions exposed as an API resulting in more productivity.
  - APIs significantly reduce development costs by reducing development efforts.
  - Improves collaboration and connectivity across the ecosystem.

APPLICATION LAYER: APIS

- Different types of APIs?
  - Representational State Transfer API (REST API)
  - Simple Object Access Protocol API (SOAP API)
  - Remote Procedure Calls (RPC)
  - WebSockets
  - GraphQL APIs

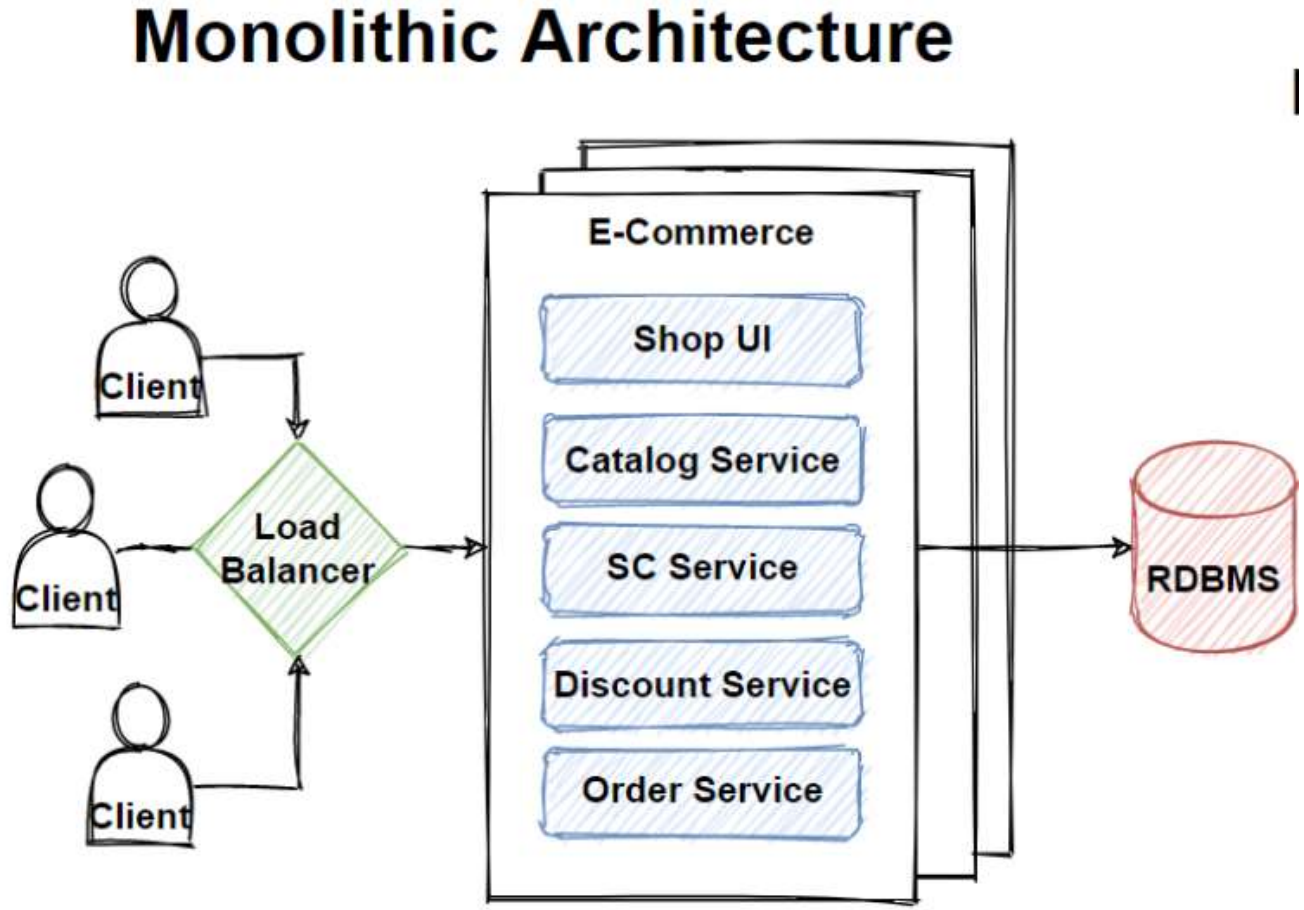
More on some of these in a separate lecture.



ARCHITECTING THE APPLICATION LAYER

- Some well known examples
  - Monolithic Architecture
  - Microservices Architecture
  - Service Oriented Architecture (SOA)
    - [Read] What is SOA?

ARCHITECTING THE APPLICATION LAYER: MONOLITHIC



Source: https://medium.com/design-microservices-architecture-with-patterns/monolithic-architecture-is-still-worth-at-2021-98bfc112dc24

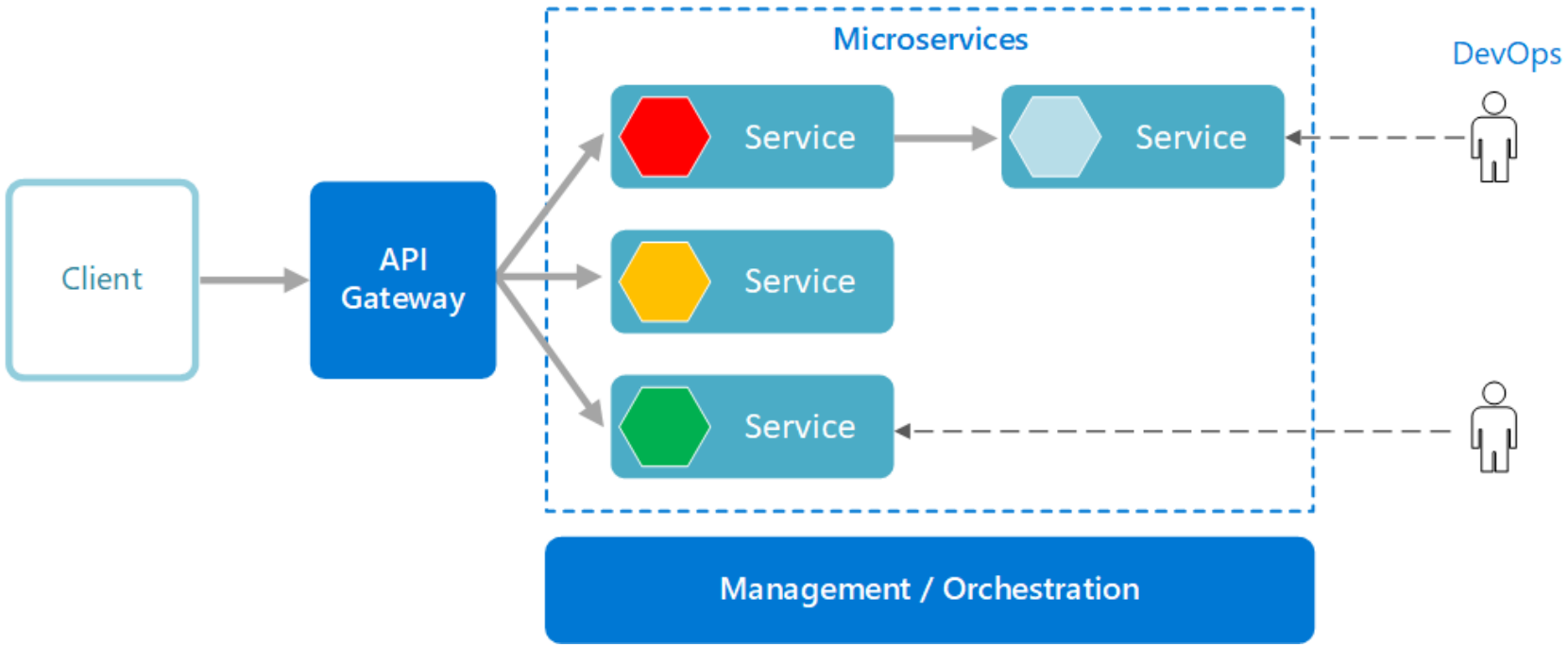
ARCHITECTING THE APPLICATION LAYER: MONOLITHIC

- Drawbacks in Monolithic Architecture?
  - Slower development speed due to **local complexity.**
  - Can't scale individual components.
  - Reliability – if there's an error in any module, it could affect the entire application's availability.
  - Barrier to technology adoption – any changes in the framework or language affects the entire application, making changes often expensive and time-consuming.
  - Lack of flexibility – a monolith is constrained by the technologies already used in the monolith.
  - A small change to a monolithic application requires the redeployment of the entire monolith.

ARCHITECTING THE APPLICATION LAYER: MICROSERVICES

- “Microservices architecture is an approach in which a single application is composed of many loosely coupled and independently deployable smaller services.” (Source – IBM)

ARCHITECTING THE APPLICATION LAYER: MICROSERVICES



Source: https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/microservices

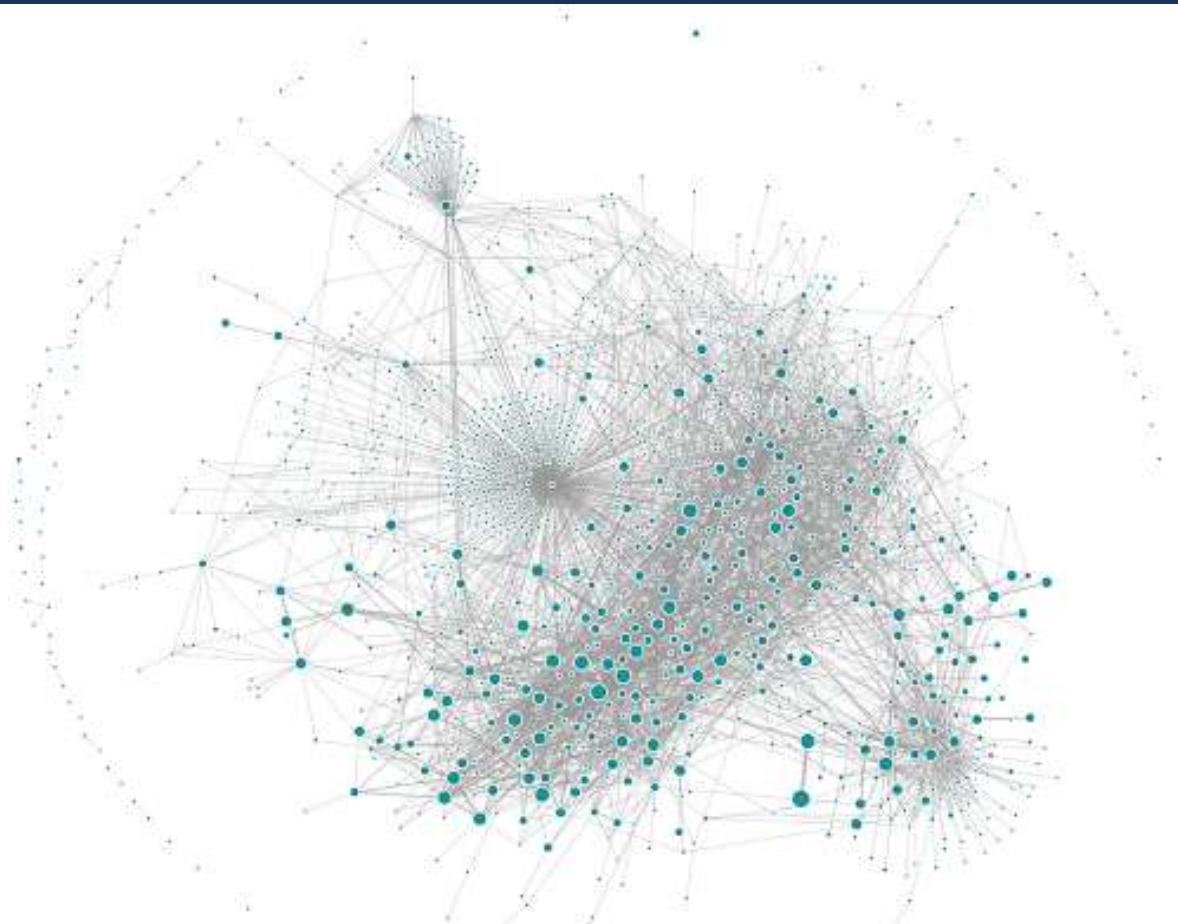
ARCHITECTING THE APPLICATION LAYER: MICROSERVICES

- Some benefits of Microservices
  - Microservices are **small, independent, and loosely coupled** (therefore, a single team can write/ maintain a single service).
  - Services can be **deployed independently** (can update an existing service without rebuilding/ redeploying the entire application).
  - Can **scale** each service independently.
  - High **reliability**.
  - Supports **polyglot programming**.

ARCHITECTING THE APPLICATION LAYER: MICROSERVICES

- Is everything great with Microservices?

ARCHITECTING THE APPLICATION LAYER: MICROSERVICES



Uber's microservice architecture circa mid-2018

Source: https://www.uber.com/en-LK/blog/microservice-architecture/

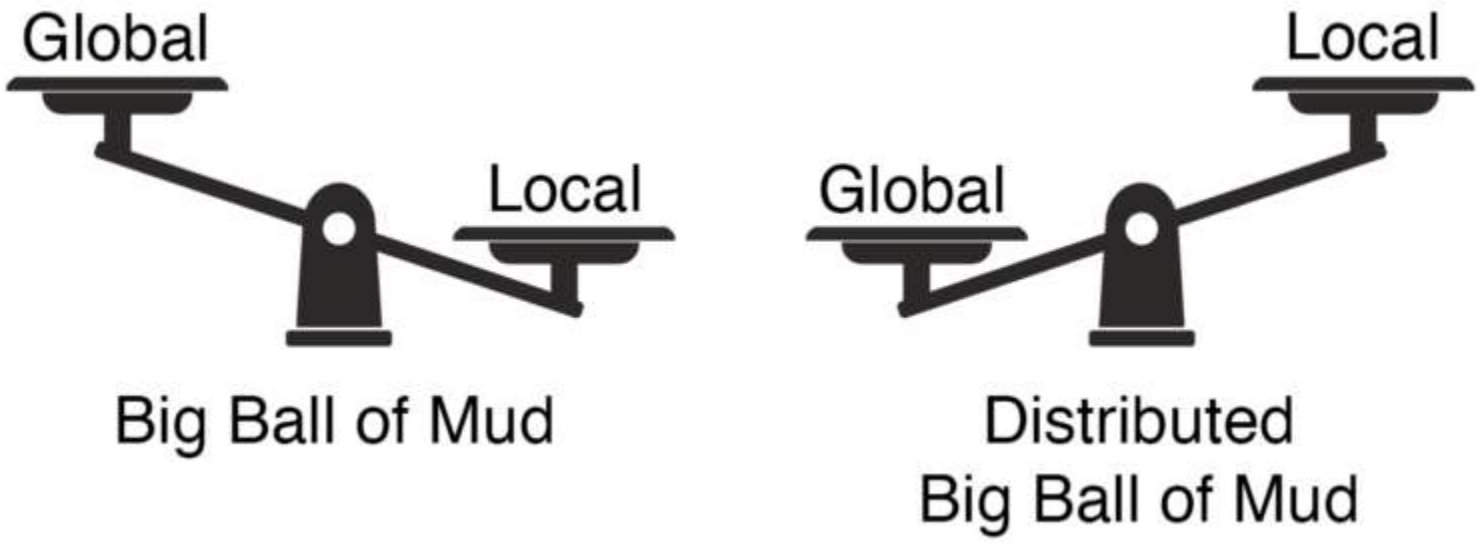
ARCHITECTING THE APPLICATION LAYER: MICROSERVICES

- **High global complexity** is the biggest problem when it comes to Microservices.
  - *Local complexity* depends on the **implementation of a service**; *global complexity* is defined by the **interactions and dependencies** between the services.
- Managing complexity is a constant challenge in Microservices.

ARCHITECTING THE APPLICATION LAYER: MICROSERVICES

- What are some other downsides of Microservices?
  - High infrastructure costs
  - Debugging challenges
  - A lot more…

BIG BALL OF MUD: AN ARCHITECTURAL ANTI-PATTERN



Source: https://www.doit.com/untangling-microservices-or-balancing-complexity-in-distributed-systems/

OTHER HELPFUL TECHNOLOGIES

- API Gateways (authentication, throttling, orchestration, caching, statistics etc.)
- Content Delivery Networks (CDN)
- Caching Tools (e.g., Redis)
- Load Balancers
- Message Queues
- Cloud Computing
  - Cloud Storage
  - Virtual Machines (VMs)
  - Many more resources…

SOME INTERESTING TOPICS

- Component Based Architecture
- Micro Frontends – Microservices for the frontend
- BFF: Backend For Frontend also known as BIF: Backend In Frontend

REFERENCES

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2. What is three-tier architecture?
3. SOAP APIs vs REST APIs
4. Micro Frontends: What are They and When to Use Them?
5. Model–view–presenter
6. An In-Depth Guide to Component-Based Architecture: Features, Benefits, and more
7. The Complete Guide To BFF (Backend For Frontend)
8. Microservices vs. monolithic architecture
9. Untangling Microservices or Balancing Complexity in Distributed Systems
10. Big Ball of Mud - The Daily Software Anti-Pattern

THANK YOU

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